

Nonstationary spawner-recruitment dynamics for Pacific salmon workshop

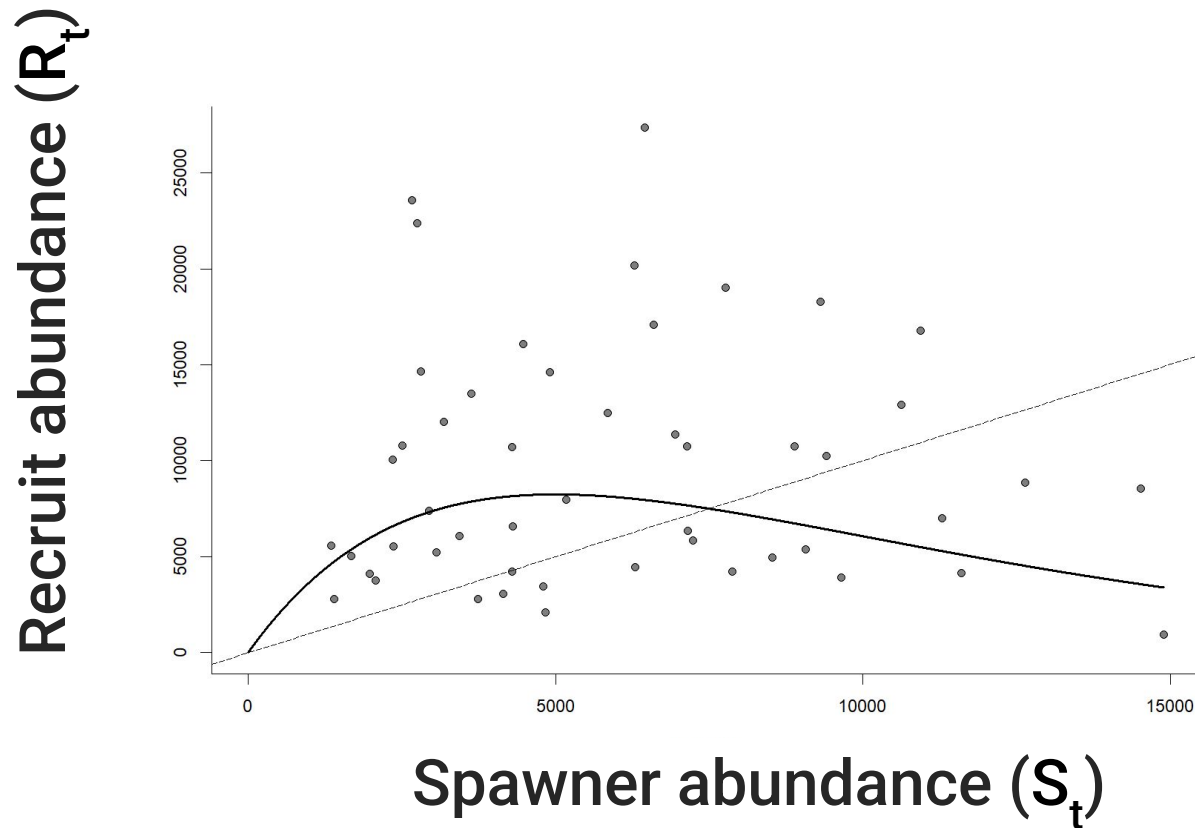


Jill Pelto, *Salmon Population Decline*, Watercolor and Colored Pencil, 2015

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Dylan Glaser**

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Fisheries and Oceans Canada**

The spawner-recruitment relationship - Ricker model

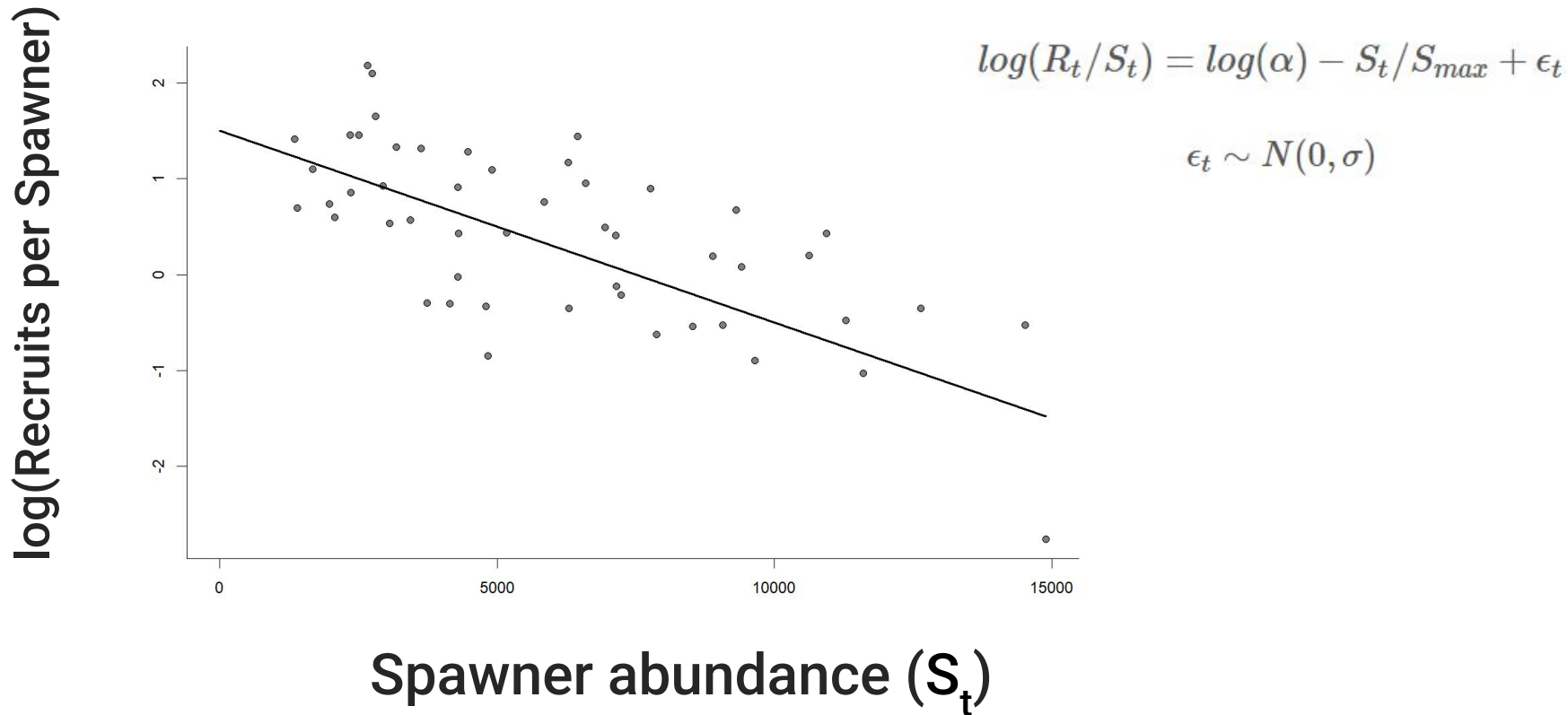


$$R_t = S_t \cdot e^{\alpha - \beta \cdot S_t + \epsilon_t}$$

$$\epsilon_t \sim N(0, \sigma)$$

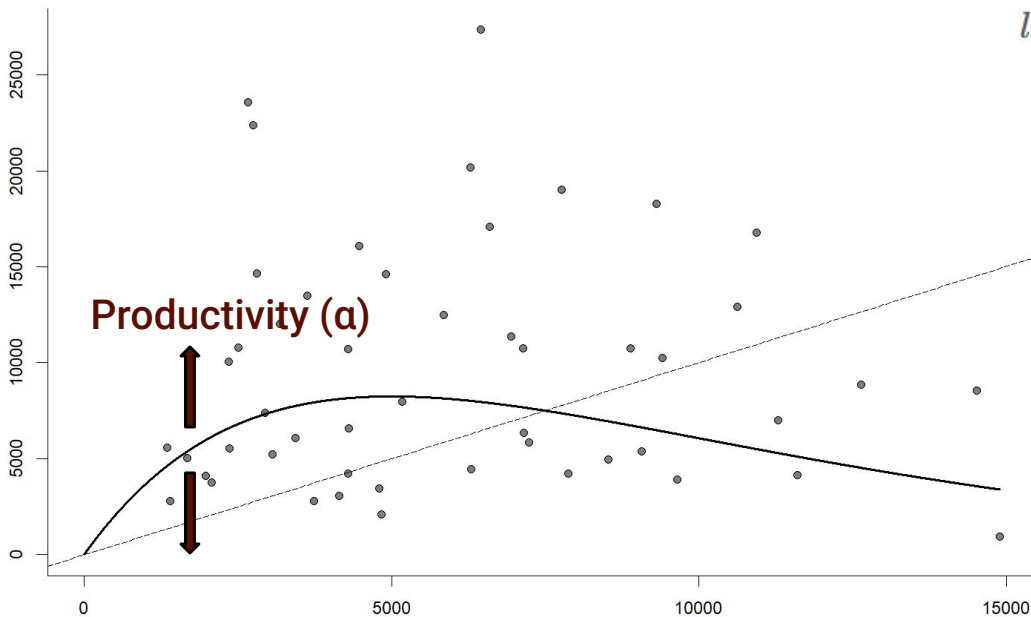
Recruits = back calculation of run size x age proportion samples to their origin year

The spawner-recruitment relationship



The spawner-recruitment relationship

Recruit abundance (R_t)



$$\log(R_t/S_t) = \log(\alpha) - S_t/S_{max} + \epsilon_t$$

$$\epsilon_t \sim N(0, \sigma)$$

Productivity - α or
 $\log(\alpha)$

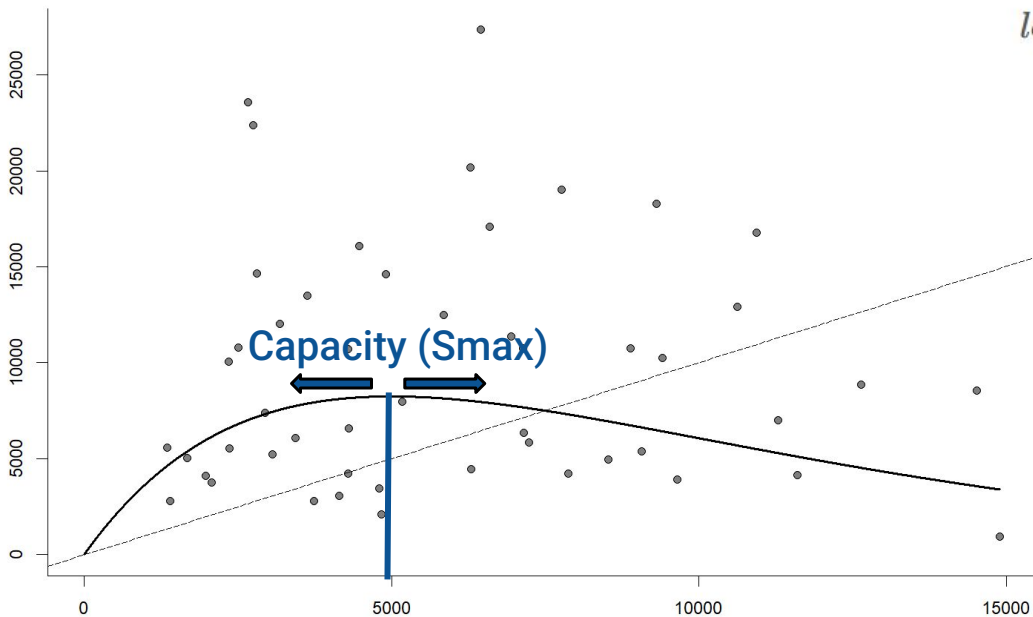
Density-independent survival from
eggs

to adult recruits

Spawner abundance (S_t)

The spawner-recruitment relationship

Recruit abundance (R_t)



$$\log(R_t/S_t) = \log(\alpha) - S_t/S_{max} + \epsilon_t$$

$$\epsilon_t \sim N(0, \sigma)$$

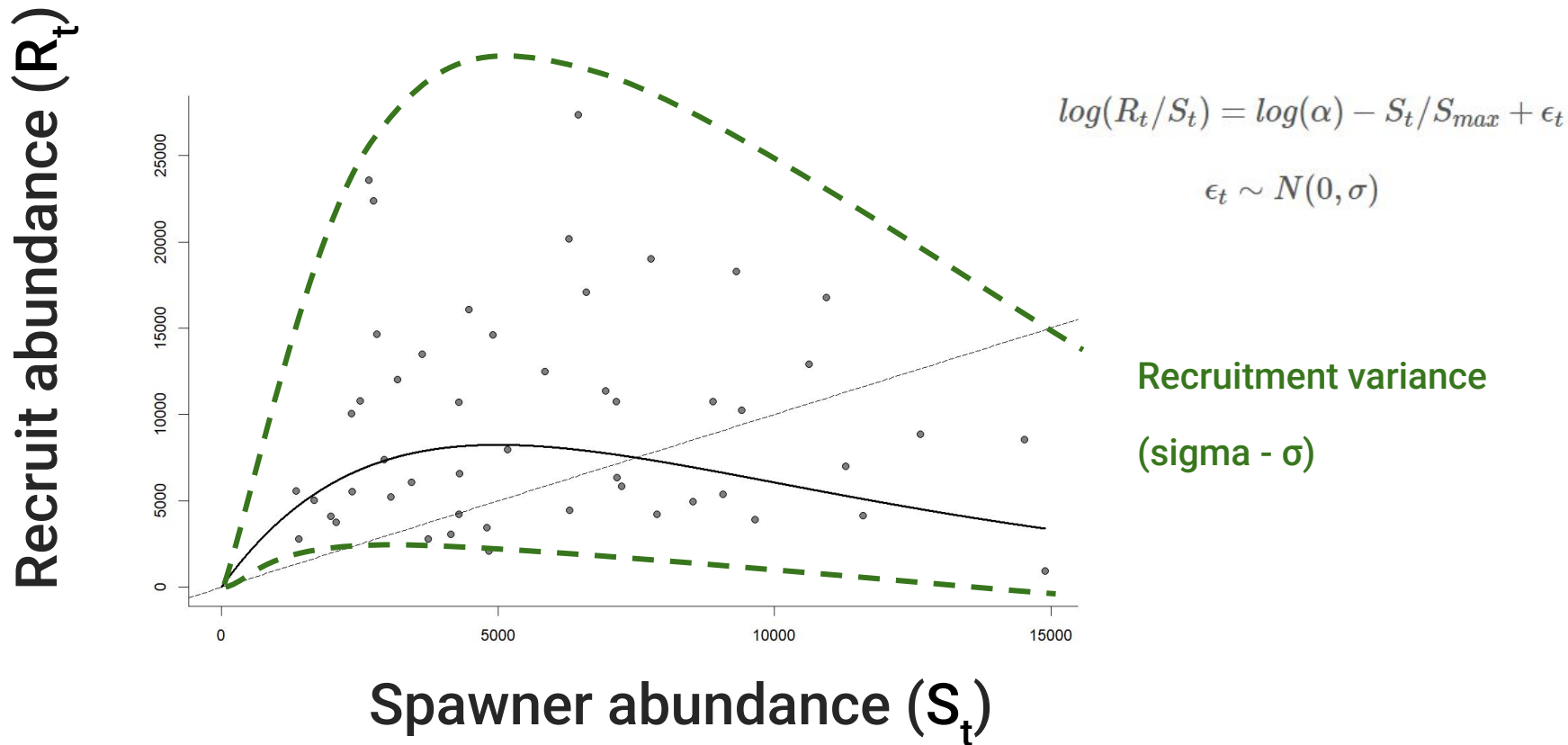
$$S_{max} = 1/\beta$$

Capacity (S_{max})

Spawning abundance that maximizes recruitment
a factor of density-dependent survival

Spawner abundance (S_t)

The spawner-recruitment relationship



Example drivers of population productivity and capacity

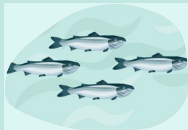
Intrinsic



Size/growth



Fecundity



Maturity/age structure

Freshwater



Thermal conditions



Habitat quality & quantity



Hydrology

Marine



Ocean climate cycles



Climate change

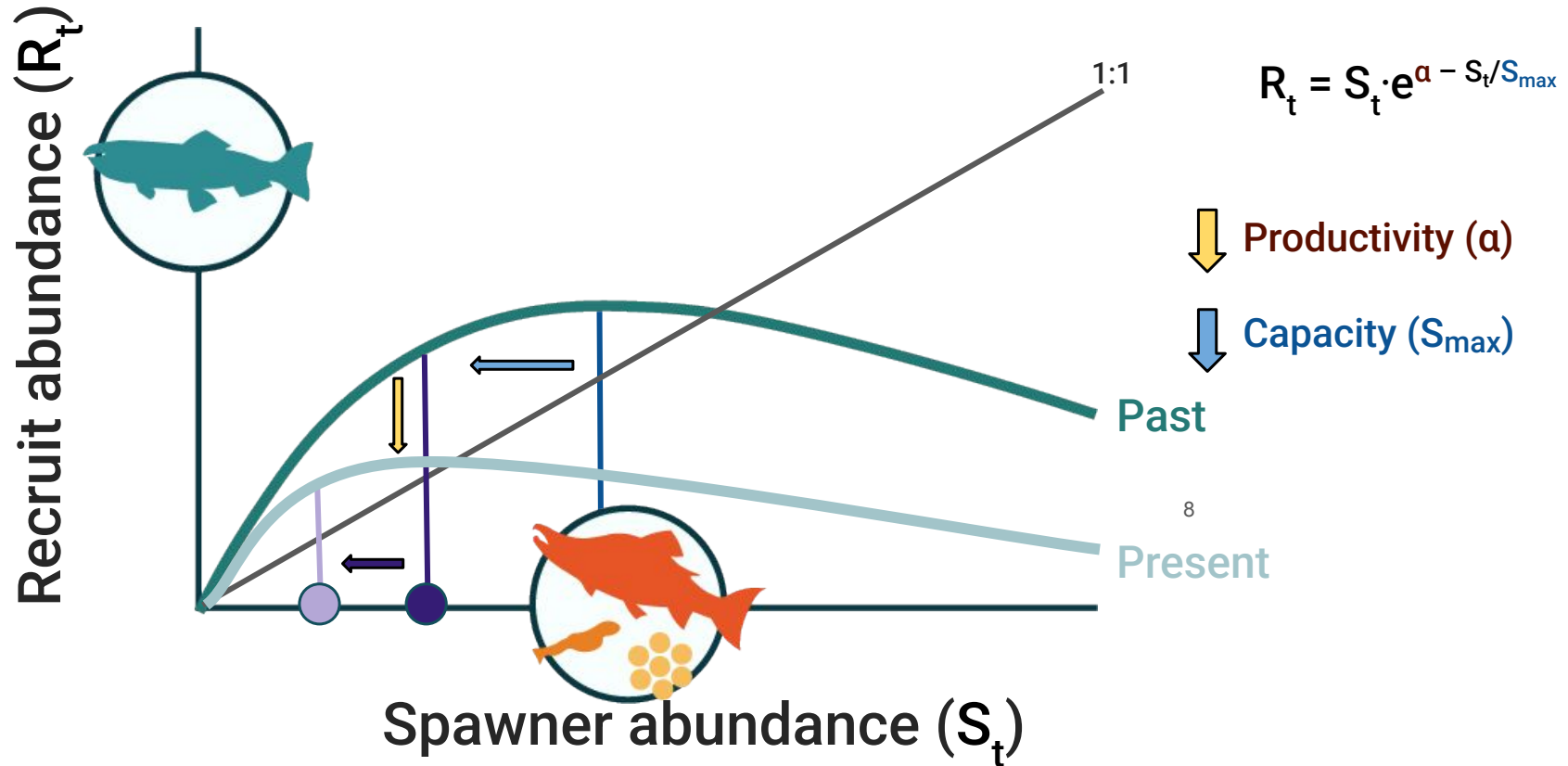


Predators & pathogens

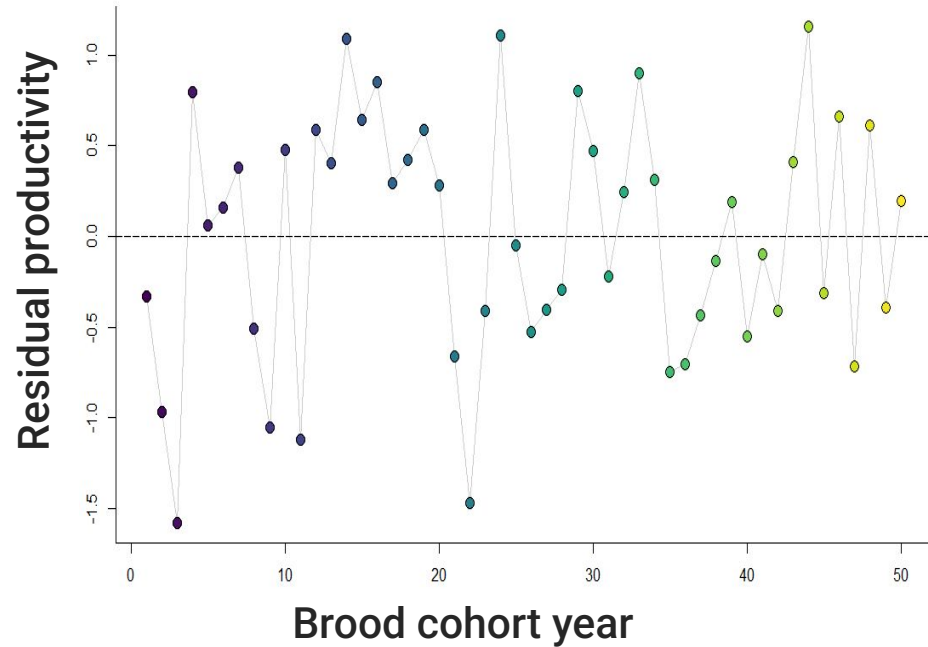
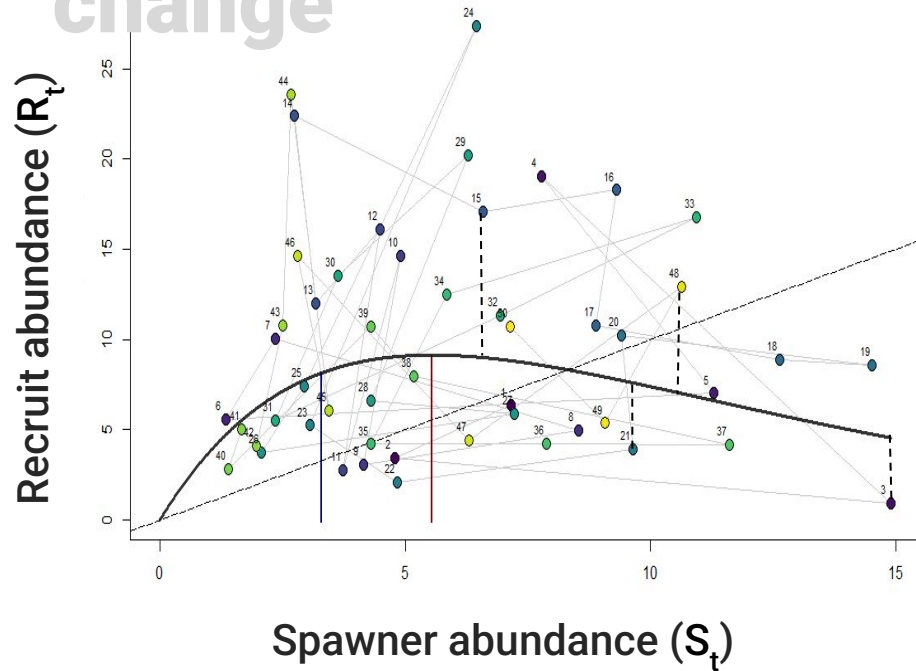


Prey quantity & quality

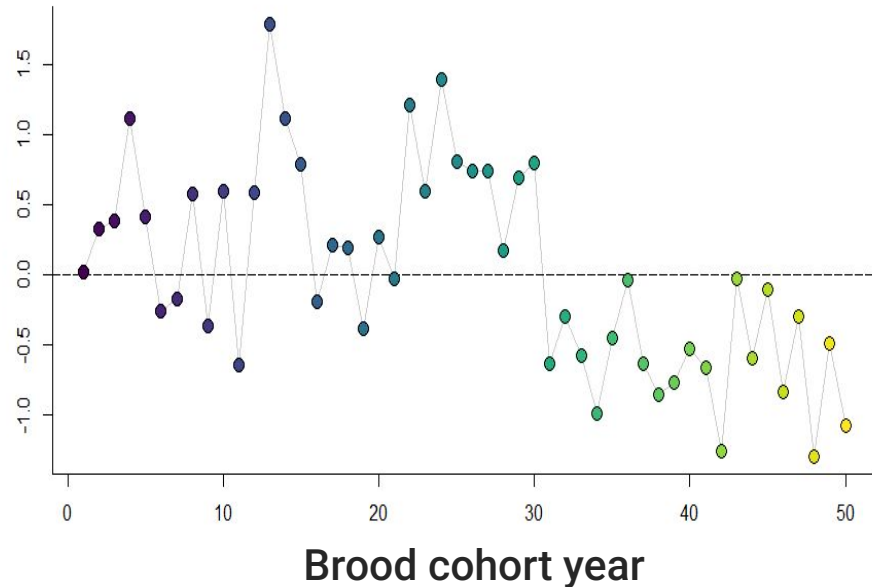
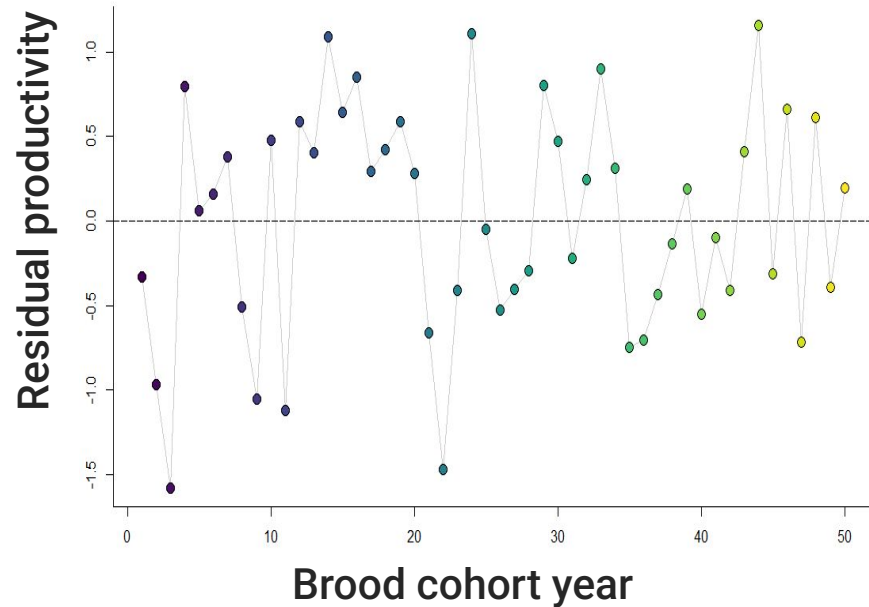
The spawner-recruitment relationship may change through time



The time-series of spawner-recruit series has information relevant to assessing change



The residual patterns are indicative of how the system may be shifting through time



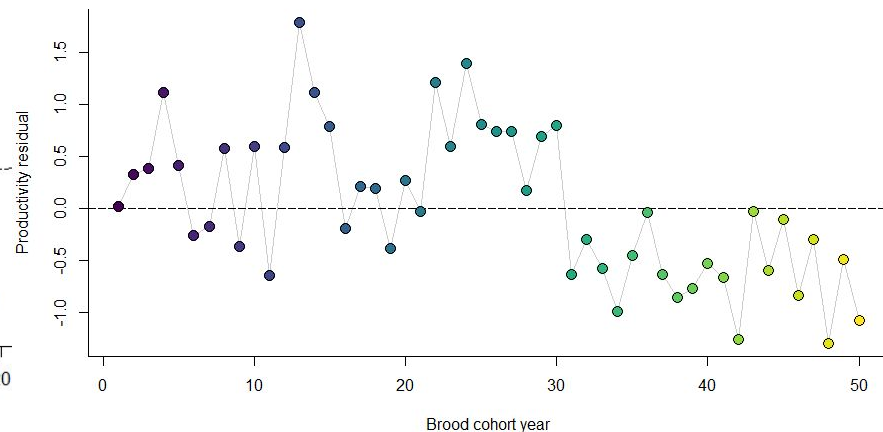
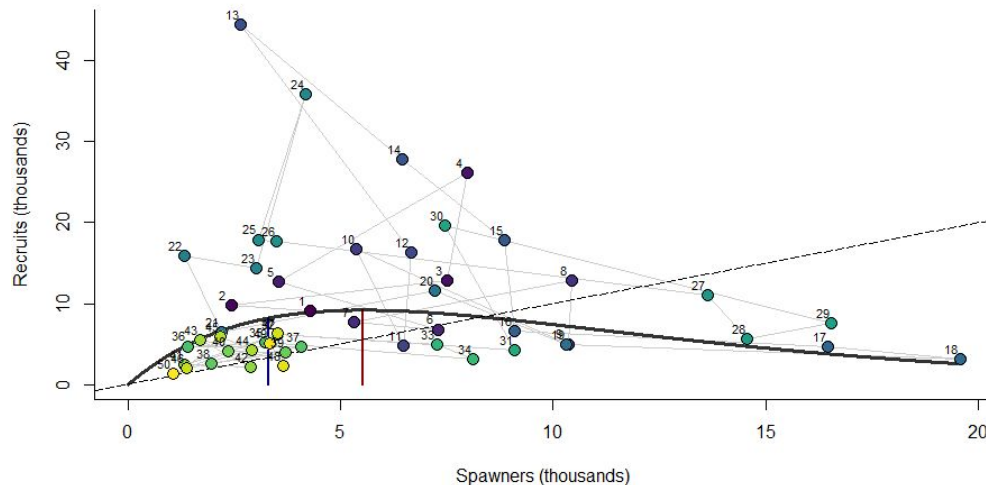
Stable Ricker model, with autocorrelated noise

$$\log(R_t/S_t) = \log(\alpha) - S_t/S_{max} + \epsilon_t$$

$$\epsilon_t = \rho \cdot \epsilon_{t-1} + \sqrt{1 - \rho^{2\Delta t}} \cdot \delta_t$$

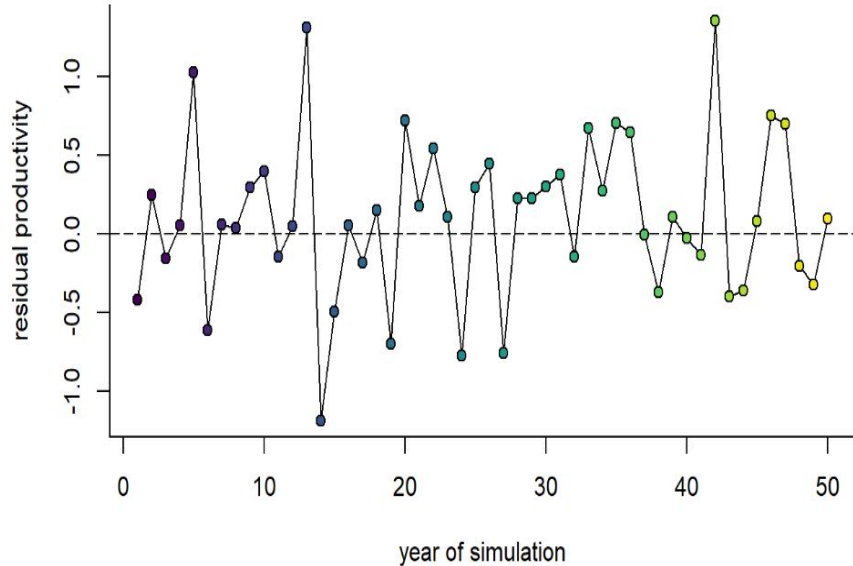
$$\delta_t \sim N(0, \sigma)$$

$$\rho = 0.6$$

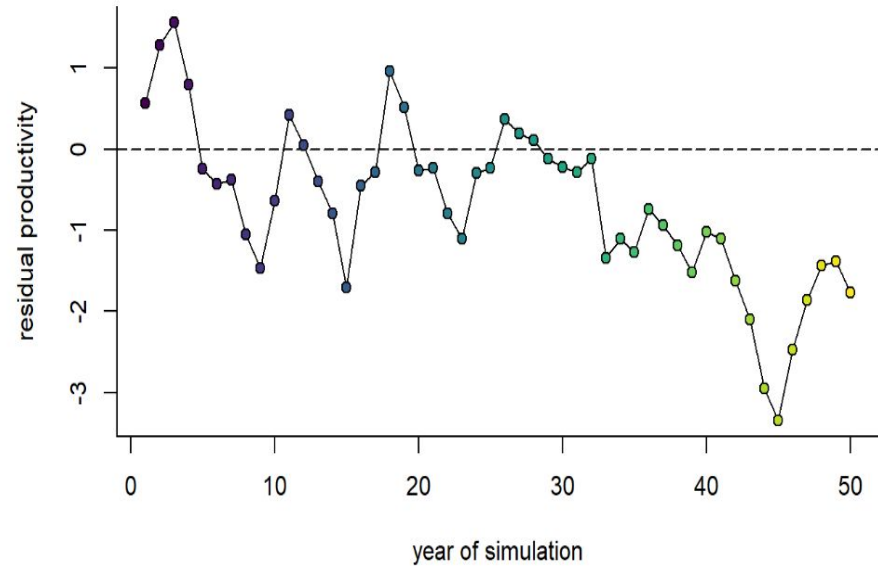


Uncorrelated and correlated fluctuations

“White” noise
(=uncorrelated)



“Red” noise
(=correlated)

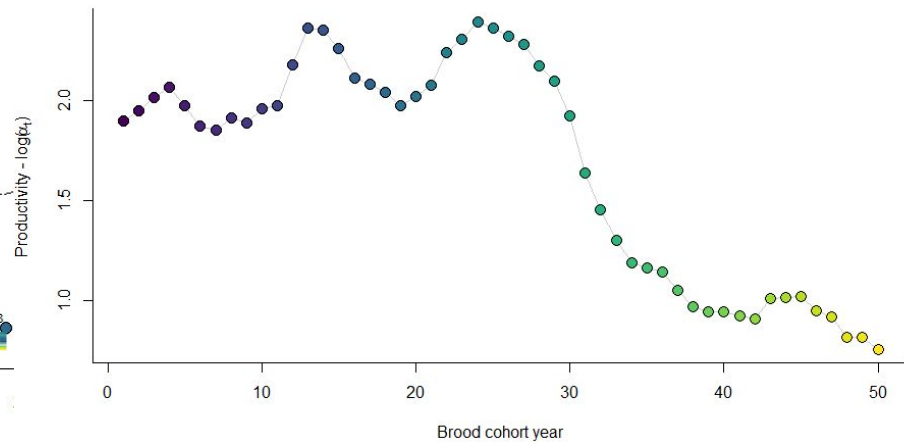
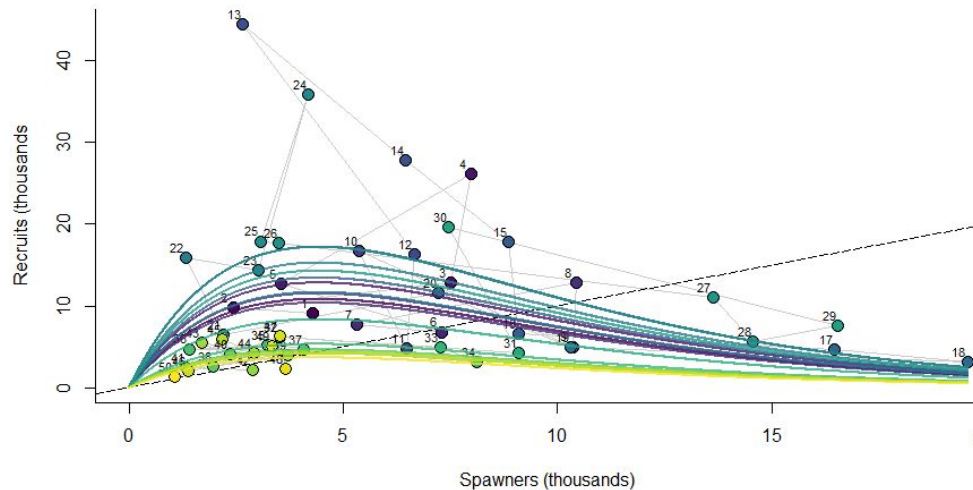


What if - there is instead continuous change in system dynamics?

$$\log(R_t/S_t) = \log(\alpha)_t - S_t/S_{max} + \epsilon_t$$

$$\log(\alpha)_t = \log(\alpha)_{t-1} + \omega_t \quad \longleftarrow \text{"Random walk"}$$

$$\omega_t \sim N(0, \sigma_\omega)$$



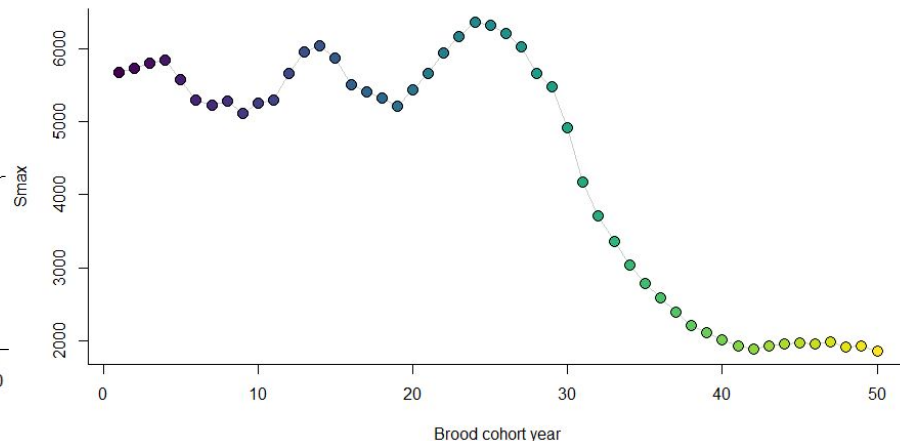
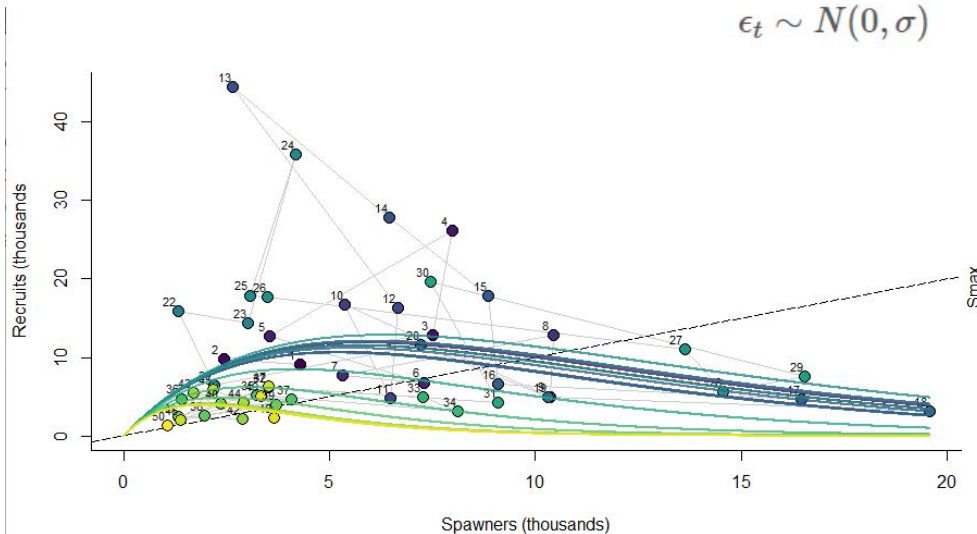
What if - there is instead continuous change in system dynamics?

$$\log(R_t/S_t) = \log(\alpha) - S_t/S_{max,t} + \epsilon_t$$

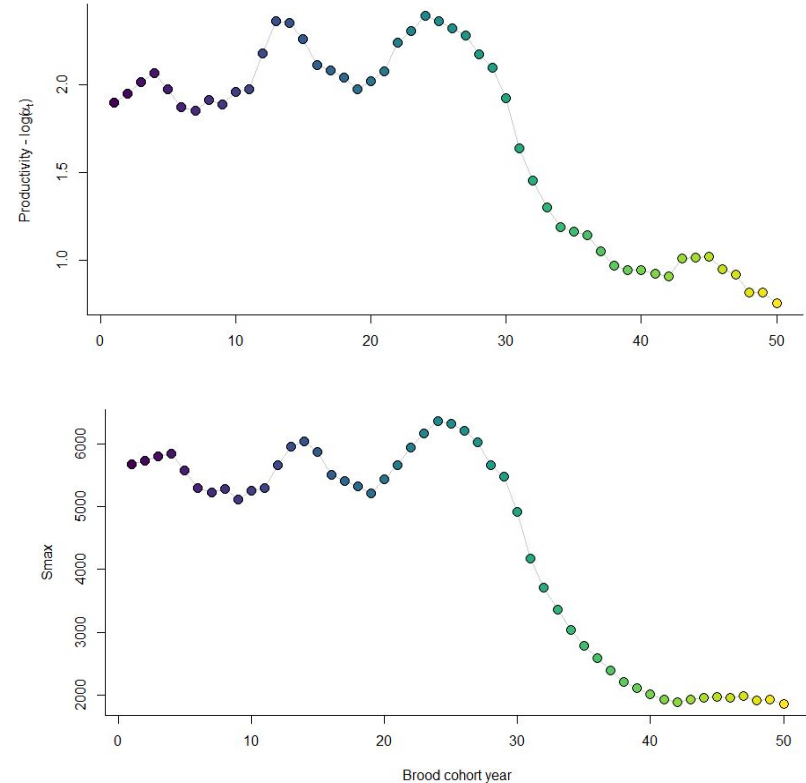
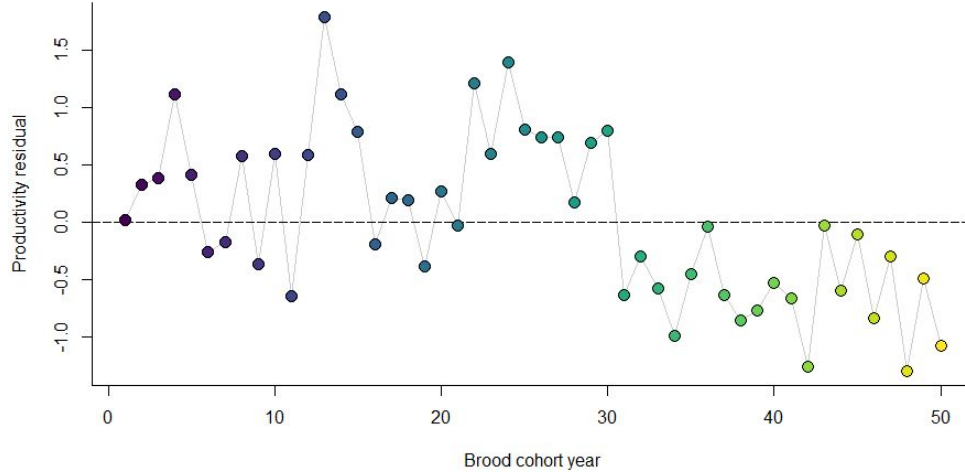
$$S_{max,t} = S_{max,t-1} + \omega_t \quad \longleftarrow \text{Random walk''}$$

$$\omega_t \sim N(0, \sigma_\omega)$$

$$\epsilon_t \sim N(0, \sigma)$$

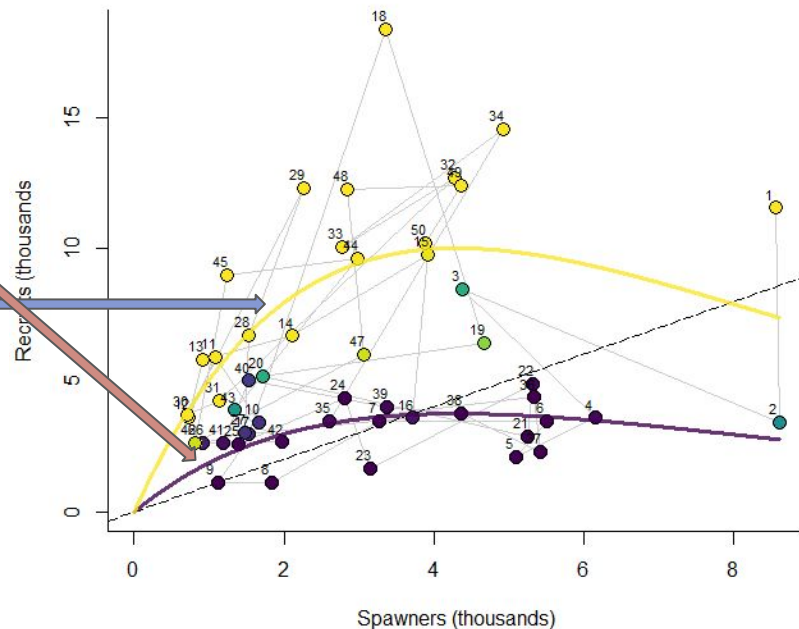
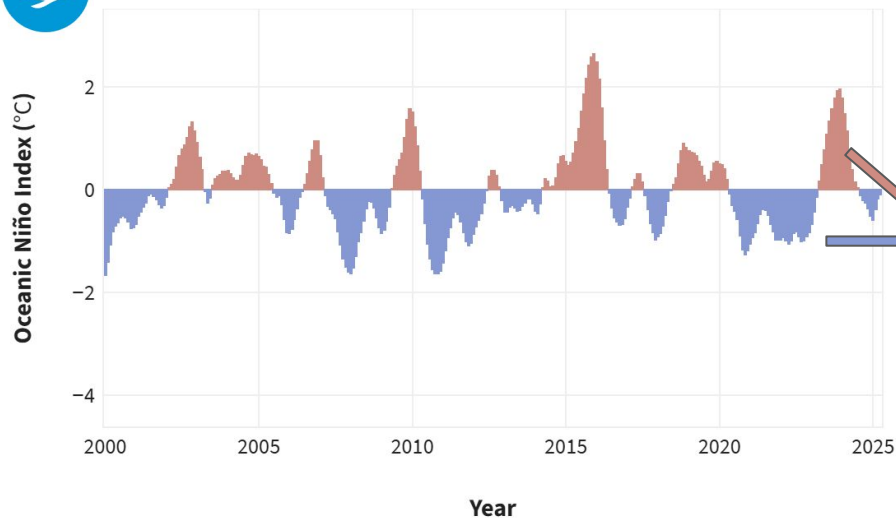


The residual patterns are indicative of how the system may be shifting through time



Regime shifts - alternative stable states

In some cases, we may not expect a single stable system - but rather multiple alternative stable regimes that a population cycles through



The Hidden Markov Ricker model

Hidden Markov models - estimate a sequence of 'hidden' states (with alternative dynamics) through time

$$\log(R_t/S_t) = \log(\alpha)_k - S_t/S_{max,k} + \epsilon_t$$

$k = 2$ regimes = 2 sets of estimated parameters

$$q_{i,j} = p(z_t = j | z_{t-1} = i)$$

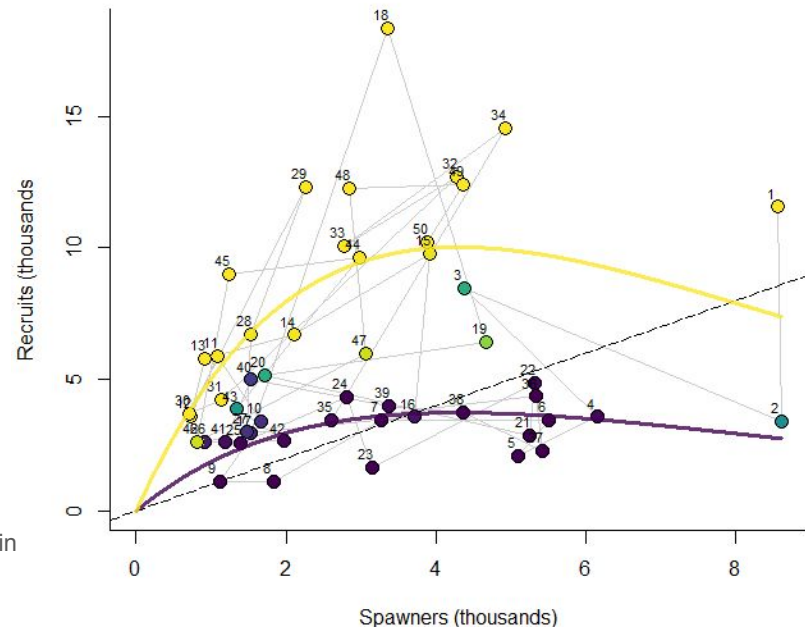
Probability of regime, depends on previous time step

$$Q = \begin{bmatrix} q_{i=1,j=1} & q_{i=1,j=2} \\ q_{i=2,j=1} & q_{i=2,j=2} \end{bmatrix}$$

Transition matrix - staying or switching regime

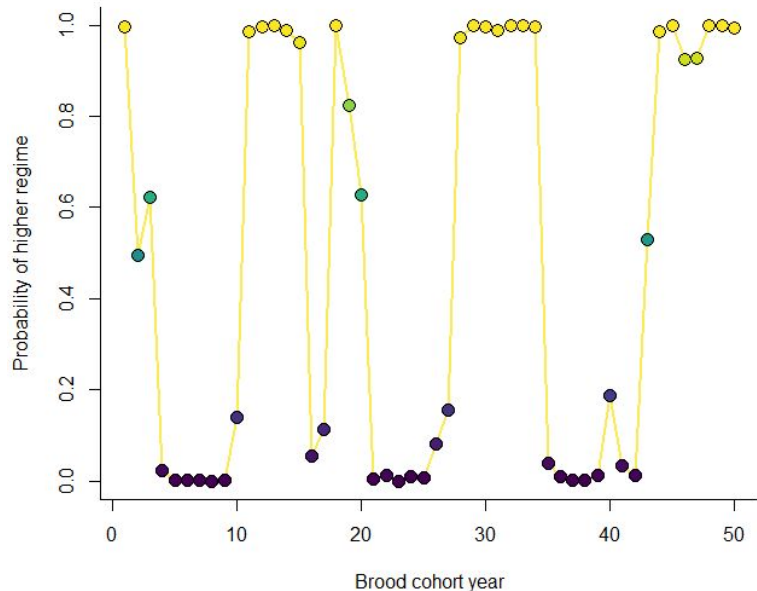
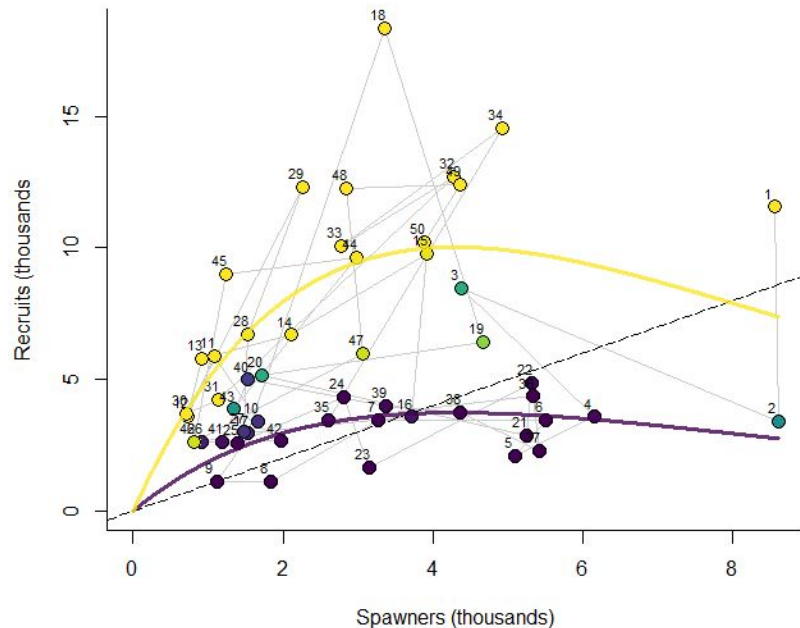
$$p(z_{1:T}, y_{1:T}) = p(z_{1:T})p(y_{1:T}|z_{1:T})$$

Likelihood: probability of a certain regime sequence, and its parameters, resulting in the observations (y)



The Hidden Markov Ricker model

Hidden Markov models - estimate a sequence of 'hidden' states (with alternative dynamics) through time



Exercise 1: Detecting stationary vs. nonstationary dynamics

Using simulation tools we've built for this workshop - you will create different scenarios of changes in key spawner-recruit parameters and estimate them with different model forms - random walk, regime shift, etc.



```
#Ricker parameters
log.a0=1.5 #initial productivity
p.change=-0.75 #proprtional change in time-varying parameter, -0.5 = -50%
              (on the log-scale), 0.5 = +50%
smax0=5000 #initial smax
sigma=0.6 #most stocks range from ~0.3 to 1.5
N=50

df.lin.prod=salmon_sim.tv(log.a0=log.a0,p.change=p.change,smax0=smax0,sig
ma=sigma,N=N,tv.par="a",tv.form="linear")
```

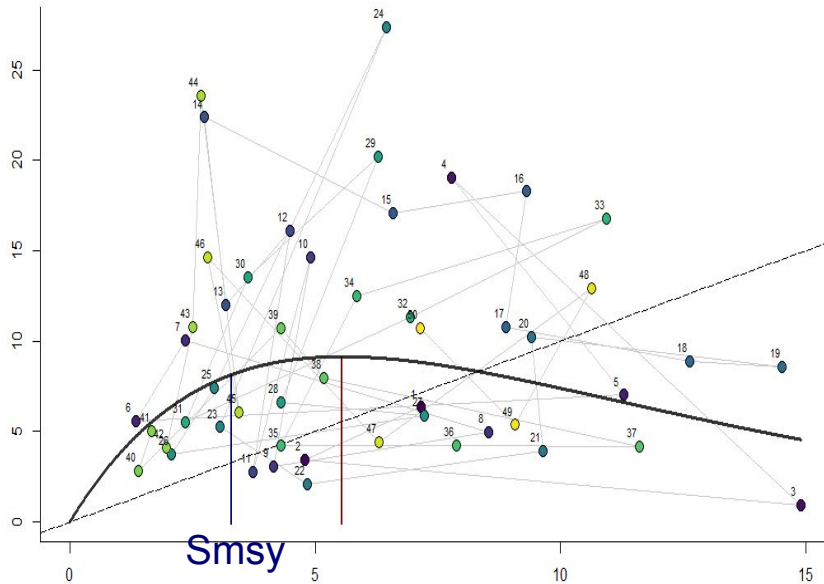
Exercise 2: Reference points and harvest control rules

$$S_{msy} = (1 - \text{LambertW}(e^{1-\log(\alpha)})) / \beta$$

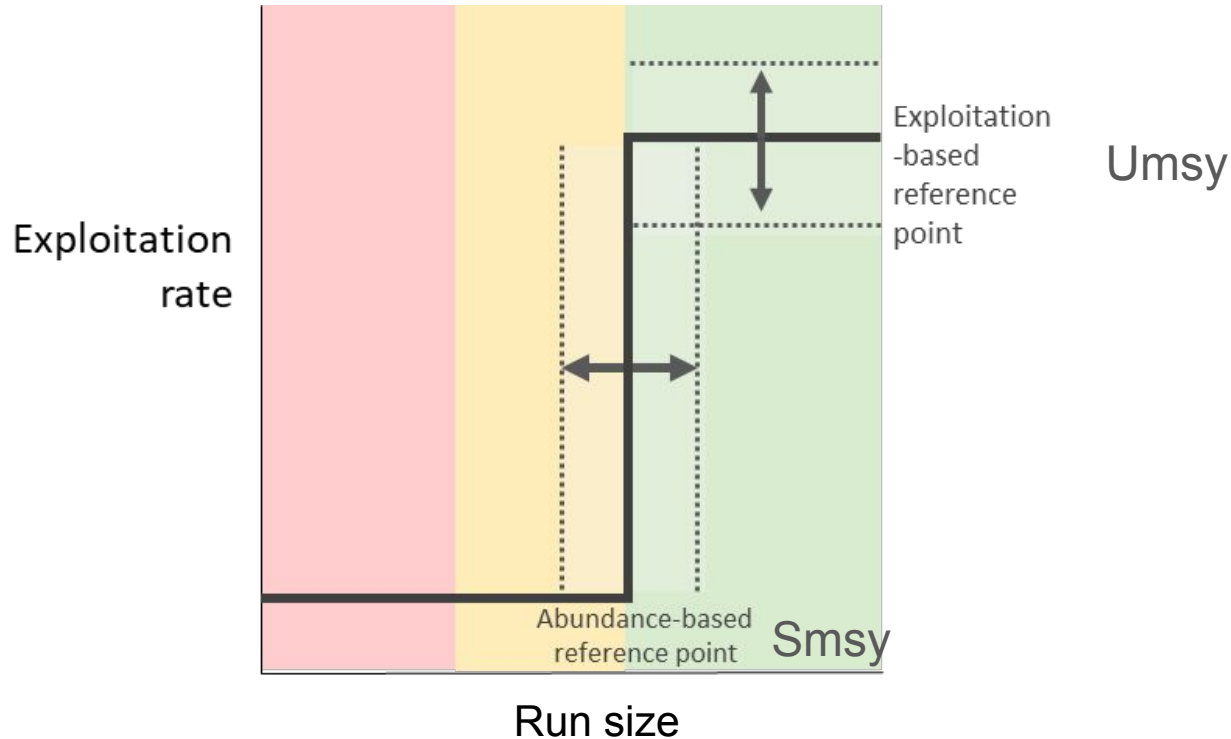
Spawners at maximum sustainable yield

$$U_{msy} = 1 - \text{LambertW}(e^{1-\log(\alpha)})$$

Harvest rate at maximum sustainable yield (0 - 1)



Exercise 2: Reference points and harvest control rules



Exercise 2: Reference points and harvest control rules

Simulation tools will now involve an internal harvest control rule with different assessment techniques (eg. static Ricker, random walk in productivity, etc.)

Can determine how choice of assessment model shapes outcomes like catch and spawner abundances

